

31.4 Estimated covariance: feasible generalized least squares

Nothing in the previous three sections is usable until $\mathbf{V}$ is known, and $\mathbf{V}$ is almost never known. The practical procedure is to choose a small family $\mathbf{V}(\boldsymbol{\theta})$, estimate $\boldsymbol{\theta}$, and proceed as if the estimate were the truth. That costs asymptotically nothing and, in small samples, a great deal — the damage falling on the standard errors rather than on the estimate.

31.4.1 The two-step estimator

Definition 31.4.1 — Feasible generalized least squares

Let $\{\mathbf{V}(\boldsymbol{\theta}) : \boldsymbol{\theta} \in \Theta\}$ be a family of positive definite $n \times n$ matrices, $\Theta \subseteq \mathbb{R}^k$, and let $\hat{\boldsymbol{\theta}}$ be an estimator of $\boldsymbol{\theta}$ computed from the data. The **feasible generalized least squares** estimator is

$$\hat{\boldsymbol{\beta}}_F = (\mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{Y}, \quad (31.4.1)$$

for $\text{rank}(\mathbf{X}) = p$, with reported covariance matrix $s_F^2 (\mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{X})^{-1}$, where s_F^2 is Eq. 31.1.6 evaluated at $\mathbf{V}(\hat{\boldsymbol{\theta}})$.

The families that matter are small. Equicorrelation inside clusters, $\mathbf{V}(\rho) = (1 - \rho)\mathbf{I} + \rho\mathbf{Z}\mathbf{Z}^\top$, and first-order autoregressive errors (Definition 21.6.1) both have $k = 1$; a variance function $\text{Var}(\varepsilon_i) = \sigma^2 \exp(\mathbf{z}_i^\top \boldsymbol{\gamma})$ has $k = \dim \boldsymbol{\gamma}$, the diagonal case treated in Section 21.3; the linear mixed models of Chapter 32 give $\mathbf{V}(\boldsymbol{\theta}) = \mathbf{Z}\mathbf{G}(\boldsymbol{\theta})\mathbf{Z}^\top + \mathbf{R}(\boldsymbol{\theta})$ with a handful of variance components. In every case k stays fixed while n grows; a family with k of the order of n cannot be estimated at all, which is why “estimate $\mathbf{V}$ freely” is not an option.

There are three routes to $\hat{\boldsymbol{\theta}}$. The **moment** route matches a few moments of the ordinary least squares residuals to the model, as in the analysis of variance estimate of an intraclass correlation used below. The **likelihood** route maximizes the profile log-likelihood of Exercise (C2), taken up in Chapter 32 with the restricted version that corrects its downward bias. The **iterative** route alternates between $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$, as in the Cochrane–Orcutt procedure of Theorem 21.6.1(d). Only consistency is needed below.

One case needs no asymptotics. If $\mathcal{C}(\mathbf{V}(\boldsymbol{\theta})\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ for every $\boldsymbol{\theta}$, the best estimator is known even though the covariance is not (Theorem 31.2.2, Exercise (B2)): in a balanced layout with random block effects, only the standard errors need $\hat{\boldsymbol{\theta}}$.

31.4.2 Large-sample equivalence

Write $\mathbf{V} = \mathbf{V}(\boldsymbol{\theta}_0)$ for the true covariance, $\hat{\boldsymbol{\beta}}_{\mathbf{V}}$ for the infeasible generalized least squares estimator that uses it, and $\mathbf{D}(\boldsymbol{\theta}) = \mathbf{V}(\boldsymbol{\theta})^{-1} - \mathbf{V}(\boldsymbol{\theta}_0)^{-1}$. The whole question is whether replacing $\boldsymbol{\theta}_0$ by $\hat{\boldsymbol{\theta}}$ disturbs the two quantities $n^{-1}\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}$ and $n^{-1/2}\mathbf{X}^\top \mathbf{V}^{-1} \boldsymbol{\varepsilon}$ that determine $\hat{\boldsymbol{\beta}}_{\mathbf{V}}$. The first is easy, because it does not involve the errors. The second is not, because $\hat{\boldsymbol{\theta}}$ is computed

from the same $\boldsymbol{\varepsilon}$.

Theorem 31.4.1 — Feasible generalized least squares

Assume Eq. 31.1.1 with $\text{rank}(\mathbf{X}) = p$ fixed, and consider a sequence of models indexed by n . Suppose:

- (G1) there are constants $0 < c \leq C < \infty$ and a neighbourhood B of $\boldsymbol{\theta}_0$ such that every eigenvalue of $\mathbf{V}(\boldsymbol{\theta})$ lies in $[c, C]$ for all $\boldsymbol{\theta} \in B$ and all n ;
- (G2) $\|\mathbf{D}(\boldsymbol{\theta})\|_2 \leq L\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|$ for $\boldsymbol{\theta} \in B$, all n , with $\|\cdot\|_2$ the spectral norm;
- (G3) the rows of $\mathbf{X}$ satisfy $\|\mathbf{x}_{(i)}\| \leq K$, and $n^{-1}\mathbf{X}^\top \mathbf{V}^{-1}\mathbf{X} \rightarrow \mathbf{A}$ for a positive definite $\mathbf{A}$;
- (G4) $\hat{\boldsymbol{\theta}} \rightarrow \boldsymbol{\theta}_0$ in probability.

Then:

- (a) $n^{-1}\mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\theta}})^{-1}\mathbf{X} \rightarrow \mathbf{A}$ in probability, so the reported covariance factor is consistent up to the scale σ^2 .
- (b) If in addition $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = O_p(1)$ and
 - (G5) $\sup_{\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\| \leq \delta} \|n^{-1/2}\mathbf{X}^\top \mathbf{D}(\boldsymbol{\theta})\boldsymbol{\varepsilon}\| = O_p(\delta)$ as $\delta \rightarrow 0$, uniformly in n ,
 then $\sqrt{n}(\hat{\boldsymbol{\beta}}_F - \hat{\boldsymbol{\beta}}_V) \rightarrow \mathbf{0}$ in probability. Consequently $\hat{\boldsymbol{\beta}}_F$ has whatever limiting distribution $\hat{\boldsymbol{\beta}}_V$ has, and in particular $\sqrt{n}(\hat{\boldsymbol{\beta}}_F - \boldsymbol{\beta}) \rightarrow N_p(\mathbf{0}, \sigma^2 \mathbf{A}^{-1})$ whenever $\sqrt{n}(\hat{\boldsymbol{\beta}}_V - \boldsymbol{\beta})$ does.

Proof.

- (a) On the event $\hat{\boldsymbol{\theta}} \in B$, whose probability tends to one by (G4),

$$\begin{aligned} \|n^{-1}\mathbf{X}^\top \mathbf{D}(\hat{\boldsymbol{\theta}})\mathbf{X}\|_2 &\leq n^{-1}\|\mathbf{D}(\hat{\boldsymbol{\theta}})\|_2\|\mathbf{X}\|_2^2 \\ &\leq n^{-1}L\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\| \cdot nK^2 = LK^2\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|, \end{aligned}$$

using $\|\mathbf{X}\|_2^2 \leq \|\mathbf{X}\|_F^2 = \sum_i \|\mathbf{x}_{(i)}\|^2 \leq nK^2$. By (G4) this tends to zero in probability, and adding $n^{-1}\mathbf{X}^\top \mathbf{V}^{-1}\mathbf{X} \rightarrow \mathbf{A}$ gives (a).

- (b) Write $\mathbf{A}_n(\boldsymbol{\theta}) = n^{-1}\mathbf{X}^\top \mathbf{V}(\boldsymbol{\theta})^{-1}\mathbf{X}$ and $\mathbf{b}_n(\boldsymbol{\theta}) = n^{-1/2}\mathbf{X}^\top \mathbf{V}(\boldsymbol{\theta})^{-1}\boldsymbol{\varepsilon}$, so that

$$\begin{aligned} \sqrt{n}(\hat{\boldsymbol{\beta}}_F - \boldsymbol{\beta}) &= \mathbf{A}_n(\hat{\boldsymbol{\theta}})^{-1}\mathbf{b}_n(\hat{\boldsymbol{\theta}}), \\ \sqrt{n}(\hat{\boldsymbol{\beta}}_V - \boldsymbol{\beta}) &= \mathbf{A}_n(\boldsymbol{\theta}_0)^{-1}\mathbf{b}_n(\boldsymbol{\theta}_0). \end{aligned}$$

By (G1) and (G3), $E\|\mathbf{b}_n(\boldsymbol{\theta}_0)\|^2 = \sigma^2 \text{tr } \mathbf{A}_n(\boldsymbol{\theta}_0) \leq \sigma^2 p K^2 / c$ is bounded, so $\mathbf{b}_n(\boldsymbol{\theta}_0) = O_p(1)$. By (G5) with $\delta_n = \|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\| = O_p(n^{-1/2})$,

$$\mathbf{b}_n(\hat{\boldsymbol{\theta}}) - \mathbf{b}_n(\boldsymbol{\theta}_0) = n^{-1/2}\mathbf{X}^\top \mathbf{D}(\hat{\boldsymbol{\theta}})\boldsymbol{\varepsilon} = O_p(n^{-1/2}) = o_p(1),$$

so $\mathbf{b}_n(\hat{\boldsymbol{\theta}}) = \mathbf{b}_n(\boldsymbol{\theta}_0) + o_p(1)$ and both are $O_p(1)$. By (a), $\mathbf{A}_n(\hat{\boldsymbol{\theta}})$ and $\mathbf{A}_n(\boldsymbol{\theta}_0)$ both converge in probability to the invertible $\mathbf{A}$, and matrix inversion is continuous there, so $\mathbf{A}_n(\hat{\boldsymbol{\theta}})^{-1} - \mathbf{A}_n(\boldsymbol{\theta}_0)^{-1} = o_p(1)$. The difference of the two displays is therefore

$$\begin{aligned} &\{\mathbf{A}_n(\hat{\boldsymbol{\theta}})^{-1} - \mathbf{A}_n(\boldsymbol{\theta}_0)^{-1}\}\mathbf{b}_n(\hat{\boldsymbol{\theta}}) \\ &\quad + \mathbf{A}_n(\boldsymbol{\theta}_0)^{-1}\{\mathbf{b}_n(\hat{\boldsymbol{\theta}}) - \mathbf{b}_n(\boldsymbol{\theta}_0)\}, \end{aligned}$$

a sum of products of $o_p(1)$ and $O_p(1)$ terms, hence $o_p(1)$. The last claim is Slutsky's theorem. ■

Condition (G5) is the substance of the theorem, and stating it as a hypothesis rather than proving it is the honest course: it separates “the weights are random” from “the weights might as well be fixed”, and it is not automatic. The naive bound $\|n^{-1/2}\mathbf{X}^\top \mathbf{D}\boldsymbol{\varepsilon}\| \leq n^{-1/2}LK\sqrt{n}\delta\|\boldsymbol{\varepsilon}\|$ is of order $\delta\sqrt{n}$, far too crude, because it ignores the cancellation among the n terms. The next lemma supplies that cancellation for one fixed $\boldsymbol{\theta}$ in the case that matters most, independent clusters of bounded size.

Lemma 31.4.2 — The drift term for independent clusters

Suppose the observations fall into G independent clusters of size at most $\bar{m}$, that $\mathbf{V}(\boldsymbol{\theta})$ is block diagonal with one block per cluster for every $\boldsymbol{\theta}$, and that (G1)–(G3) hold. Then for every fixed $\boldsymbol{\theta} \in B$,

$$\mathbb{E}\|n^{-1/2}\mathbf{X}^\top \mathbf{D}(\boldsymbol{\theta})\boldsymbol{\varepsilon}\|^2 \leq \sigma^2 \bar{m} K^2 L^2 C \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|^2.$$

In particular $n^{-1/2}\mathbf{X}^\top \mathbf{D}(\boldsymbol{\theta})\boldsymbol{\varepsilon} = O_p(\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|)$.

Proof. Write $\mathbf{X}_k$, $\boldsymbol{\varepsilon}_k$ and $\mathbf{D}_k$ for the blocks belonging to cluster k , so that $\mathbf{X}^\top \mathbf{D}\boldsymbol{\varepsilon} = \sum_k \mathbf{X}_k^\top \mathbf{D}_k \boldsymbol{\varepsilon}_k$. The summands are independent with mean zero, so their covariance matrices add and

$$\mathbb{E}\|n^{-1/2}\mathbf{X}^\top \mathbf{D}\boldsymbol{\varepsilon}\|^2 = \frac{1}{n} \sum_k \mathbb{E}\|\mathbf{X}_k^\top \mathbf{D}_k \boldsymbol{\varepsilon}_k\|^2 \leq \frac{1}{n} \sum_k \|\mathbf{X}_k\|_2^2 \|\mathbf{D}_k\|_2^2 \mathbb{E}\|\boldsymbol{\varepsilon}_k\|^2.$$

Now $\|\mathbf{X}_k\|_2^2 \leq m_k K^2$, $\|\mathbf{D}_k\|_2 \leq \|\mathbf{D}\|_2 \leq L\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|$ by (G2), and $\mathbb{E}\|\boldsymbol{\varepsilon}_k\|^2 = \sigma^2 \text{tr } \mathbf{V}_{kk} \leq \sigma^2 m_k C$ by (G1). Since $m_k \leq \bar{m}$ and $\sum_k m_k = n$, the sum is at most $\sigma^2 \bar{m} K^2 L^2 C \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|^2$. Chebyshev's inequality turns the moment bound into the stochastic order. ■

Remark. Lemma 31.4.2 gives the right order for each fixed $\boldsymbol{\theta}$; (G5) asks for it *uniformly* over a shrinking ball, which is what allows $\boldsymbol{\theta}$ to be replaced by the data-dependent $\hat{\boldsymbol{\theta}}$. That passage is a stochastic equicontinuity argument we do not carry out: with $\boldsymbol{\theta}$ of fixed dimension, a chaining bound over the ball converts the pointwise bound into the uniform one under mild smoothness of $\boldsymbol{\theta} \mapsto \mathbf{V}(\boldsymbol{\theta})^{-1}$. Amemiya (1985, ch. 6) proves the conclusion of Theorem 31.4.1(b) for several concrete families, and Carroll and Ruppert (1988, ch. 3) for smooth variance functions; (G5) is verified there, not assumed away.

31.4.3 What goes wrong in small samples

Theorem 31.4.1 says that $\hat{\boldsymbol{\beta}}_F$ and $\hat{\boldsymbol{\beta}}_V$ are indistinguishable in large samples. It says nothing about finite n , where three things go wrong.

The estimator is no longer linear. $\hat{\boldsymbol{\beta}}_F$ is a ratio of random quantities, so none of the exact

theory of Theorem 31.1.1 applies. A symmetry argument rescues unbiasedness when $\hat{\boldsymbol{\theta}}$ is an even function of the residuals and the errors are symmetric, exactly as in Theorem 21.3.1(c), whose argument goes through verbatim for a non-diagonal $\mathbf{V}(\hat{\boldsymbol{\theta}})^{-1}$.

The variance exceeds the generalized least squares variance. Estimating $\boldsymbol{\theta}$ adds noise, with an excess of order $1/G$ in the cluster setting; Kackar and Harville (1984) give the standard expansion of it.

The reported standard error ignores that noise. The matrix $s_F^2(\mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{X})^{-1}$ is the covariance $\hat{\boldsymbol{\beta}}_{\mathbf{V}}$ would have had if $\hat{\boldsymbol{\theta}}$ were the truth: it omits the extra variability, and is evaluated at a random $\hat{\boldsymbol{\theta}}$ drawn towards values that make the fit look good. Tests are liberal and intervals too short.

Example. (How many clusters are enough?)

Simulate clusters of sizes 2, 4, 8 repeated 1, 2, 4, 8, 16, 32 times, with a cluster-level regressor z , a within-cluster regressor w , and equicorrelation $\rho = 0.6$ inside clusters. Estimate ρ by the analysis of variance moment estimator from the ordinary least squares residuals and form $\hat{\boldsymbol{\beta}}_F$. Figure 31.4.1 shows 20,000 replications for each size.

The efficiency story is almost trivial. Ordinary least squares attains about 0.895 of the optimal precision for the coefficient of z — its variance is 1.12 times the optimal one — at every number of clusters, while feasible generalized least squares already reaches 0.987 with 3 clusters and 1.000 with 96. Estimating one smooth parameter costs almost nothing in efficiency, in sharp contrast with Example (Weights from a few replicates), where a separate variance was estimated in each of ten groups.

The inference story is the opposite. The nominal 95% interval based on $s_F^2(\mathbf{X}^\top \mathbf{V}(\hat{\rho})^{-1} \mathbf{X})^{-1}$ covers the truth in 0.728 of samples with 3 clusters, 0.916 with 12, and 0.944 with 96, while the interval that uses the true ρ covers 0.950 at three clusters and stays within simulation error of 95% at every size, as Theorem 31.1.1 guarantees. The cluster-robust interval built on $\hat{\boldsymbol{\beta}}_F$ is uniformly a little better than the model-based one, but no cure when clusters are few — 0.764 at 3 — and the two become indistinguishable once there are several dozen, neither reaching the nominal level even at 96, where both stand at 0.944.

Idea. Estimating a covariance parameter buys back the efficiency almost at once and the correct standard error only slowly: efficiency depends on $\hat{\boldsymbol{\theta}}$ through a function whose derivative vanishes at the optimum, so a mediocre $\hat{\boldsymbol{\theta}}$ still gives nearly optimal weights, while the reported variance depends on $\hat{\boldsymbol{\theta}}$ to first order and inherits all its error.

31.4.4 Which repair to use

Ordinary least squares with robust standard errors assumes nothing about the shape of $\mathbf{V}$, loses the few per cent bounded by Proposition 31.2.3, and needs many clusters. *Feasible generalized least squares* recovers that efficiency, but only as far as $\mathbf{V}(\boldsymbol{\theta})$ is right; a wrong shape leaves the estimate unbiased — by the argument of Theorem 21.3.1(b), applied with $\mathbf{W} = \mathbf{V}(\boldsymbol{\theta}_1)^{-1}$ for the limiting $\boldsymbol{\theta}_1$ — and the reported covariance wrong in an unknown direction. *Both*, a fitted $\mathbf{V}(\hat{\boldsymbol{\theta}})$ with a robust standard error, is the modern default for clustered data, and Chapter 33 develops it for longitudinal designs.

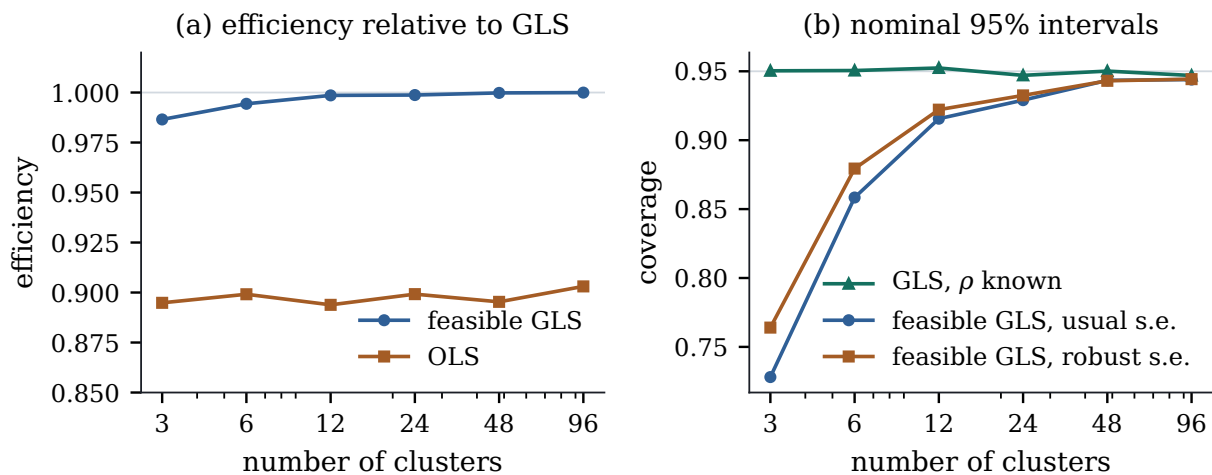


Figure 31.4.1: Feasible generalized least squares with equicorrelated clusters of sizes 2, 4, 8 and $\rho = 0.6$, for the coefficient of a cluster-level regressor (20,000 replications each). (a) Efficiency: the variance of generalized least squares with ρ known divided by the estimator's own variance. Feasible generalized least squares recovers essentially all of it at once. (b) Coverage of nominal 95% intervals: both intervals need dozens of clusters, and the cluster-robust one is no cure.

None of the three helps when the independent units are few: with three clusters $\hat{\rho}$ is hopeless and a sandwich with three terms is not a covariance estimate. The remedy is a better small-sample approximation — the Satterthwaite and Kenward–Roger degrees of freedom of [Chapter 32](#), or the bias-corrected sandwiches and cluster bootstrap of [Chapter 23](#) — or more clusters.

Example. (The El Niño trend without assuming ρ)

The analysis of Example (A trend in sea surface temperature) assumed $\rho = 0.5$. The moment estimator from the ordinary least squares residuals gives $\hat{\rho} = 0.610$, and the feasible generalized least squares fit then reports a trend of 0.1346 degrees per decade with standard error 0.0618. For comparison, the ordinary least squares trend, 0.1349, carries the usual standard error 0.0223 and a cluster-robust standard error of 0.0584 with years as clusters. The two honest routes agree closely and the naive one is out by a factor of nearly three. With 61 clusters both are in the range where [Example \(How many clusters are enough?\)](#) found them reliable.

```
import numpy as np

def quasi_demean(A, Z, sizes, rho):
    """Whitening for equicorrelation rho: rows of A are vectors in R^n."""
    c = 1.0 - np.sqrt((1 - rho) / (1 - rho + sizes * rho))
    return (A - (A @ Z / sizes * c) @ Z.T) / np.sqrt(1 - rho)

def fit(X, Y, Z, sizes, rho):
    """GLS coefficients and the model-based covariance factor, for many data sets at once."""
    Xw = quasi_demean(X.T, Z, sizes, rho).T
    Yw = quasi_demean(Y, Z, sizes, rho)
    A = np.linalg.inv(Xw.T @ Xw)
    B = Yw @ Xw @ A
    R = Yw - B @ Xw.T
```

```

s2 = np.sum(R * R, axis=1) / (X.shape[0] - X.shape[1])
return B, A, s2

def moment_rho(R, Z, sizes):
    """ANOVA estimate of the intraclass correlation from residual vectors R."""
    G, n = len(sizes), R.shape[1]
    means = R @ Z / sizes
    between = np.sum(means**2 * sizes, axis=1) / G
    within = (np.sum(R * R, axis=1) - np.sum(means**2 * sizes, axis=1)) / (n - G)
    mbar = (n - np.sum(sizes**2) / n) / (G - 1)
    est = (between - within) / (between + (mbar - 1) * within)
    return np.clip(est, 0.0, 0.95)

```

[Run this code online](#)

```

import numpy as np
import statsmodels.api as sm

frame = sm.datasets.elnino.load_pandas().data
temps = frame.iloc[:, 1:].to_numpy()
G, m = temps.shape
n = G * m
y = temps.ravel()
month = np.tile(np.arange(1, m + 1), G)
time = (np.repeat(frame["YEAR"].to_numpy(), m) - 1980.0) + (month - 6.5) / 12.0
cols = [np.ones(n), time / 10.0]
for k in (1, 2, 3):
    cols += [np.cos(2 * np.pi * k * month / m), np.sin(2 * np.pi * k * month / m)]
X = np.column_stack(cols)
Zn = np.zeros((n, G))
Zn[np.arange(n), np.repeat(np.arange(G), m)] = 1.0
sizes_n = np.full(G, m)

b_ols = np.linalg.solve(X.T @ X, X.T @ y)
rho_hat = float(moment_rho(y - X @ b_ols)[None, :], Zn, sizes_n)[0]
b_f, A_f, s2_f = fit(X, y[None, :], Zn, sizes_n, rho_hat)
se_f = np.sqrt(s2_f[0] * A_f[1, 1])
print(f"El Nino: rho_hat = {rho_hat:.3f}, trend {b_f[0, 1]:.4f} (s.e. {se_f:.4f})")

```

[Run this code online](#)

31.4.5 Exercises

31.4.5.1 A. Check your understanding

Exercise A1.

Explain why $\hat{\beta}_F$ is not a linear function of $\mathbf{Y}$, and why this alone makes Corollary 7.2.3 inapplicable to it. Which part of Theorem 31.1.1 could still hold exactly if $\hat{\theta}$ happened to be independent of $\mathbf{Y}$?

Exercise A2.

Give a family $\mathbf{V}(\boldsymbol{\theta})$ with $k = 2$ for which no estimation of $\boldsymbol{\theta}$ is needed to obtain the best estimator of $\mathbf{X}\boldsymbol{\beta}$, and say what $\boldsymbol{\theta}$ is still needed for.

31.4.5.2 B. Practice

Exercise B1.

For balanced clusters of size m with equicorrelation ρ , let B and W be the between- and within-cluster mean squares of the ordinary least squares residuals. Show that the moment estimator $\hat{\rho} = (B - W)/\{B + (m - 1)W\}$ is the solution of matching $E B$ and $E W$ to their model values when the model is fitted without regressors, and explain why $\hat{\rho}$ can be negative. What does truncating it at zero do to [Theorem 31.4.1](#)?

Solution. For a one-way layout with no regressors, $E W = \sigma^2(1 - \rho)$ and $E B = \sigma^2\{1 + (m - 1)\rho\}$ (this is Exercise (B1)). Solving $B = \sigma^2\{1 + (m - 1)\rho\}$, $W = \sigma^2(1 - \rho)$ for ρ gives the stated ratio. Since B and W are independent chi-squared multiples, $B < W$ happens with positive probability whenever ρ is small, so $\hat{\rho} < 0$ is possible; [Chapter 32](#) discusses negative variance-component estimates at length. Truncation leaves $\hat{\rho}$ consistent whenever $\rho > 0$, which is all (G4) requires; at $\rho = 0$ the truncated estimator is still consistent and still satisfies $\sqrt{n} \hat{\rho} = O_p(1)$, so [Theorem 31.4.1\(b\)](#) still applies to $\hat{\boldsymbol{\beta}}_F$, but $\hat{\rho}$ itself is no longer asymptotically normal and the usual Wald theory for it fails.

Exercise B2.

Suppose $\mathbf{V}(\boldsymbol{\theta}) = \text{diag}(1 + \theta z_1, \dots, 1 + \theta z_n)$ with known $z_i \geq 0$ and $\theta \geq 0$. Verify (G1) and (G2) on a neighbourhood of an interior θ_0 , giving explicit constants, and show that (G3) holds when the z_i and the rows of $\mathbf{X}$ are bounded.

Exercise B3.

In the balanced cluster model of Exercise (B4), all clusters of size m , consider estimating μ with weights built from a wrong value ρ_1 when the truth is ρ_0 . Show that the resulting estimator is $\bar{Y}$ for every ρ_1 , and hence that the efficiency is exactly 1. Contrast with the unbalanced case and explain the flat curve in [Figure 31.4.1\(a\)](#).

Solution. With equal sizes, Eq. 31.1.8 gives $w_k = m/(1 - \rho_1 + m\rho_1)$, the same for every k , so the weighted mean of the cluster means is $\bar{Y}$ whatever ρ_1 is: the estimator does not depend on the weights at all. With unequal sizes the weights do depend on ρ_1 , but only through the smooth ratio $m_k/(1 - \rho_1 + m_k\rho_1)$, and the variance of the resulting estimator is stationary at $\rho_1 = \rho_0$ because ρ_0 minimizes it. A first-order error in $\hat{\rho}$ therefore produces only a second-order loss of efficiency, which is why panel (a) is flat.

31.4.5.3 C. Going deeper

Exercise C1.

Show that the bound of [Lemma 31.4.2](#) cannot be improved in order: construct a family with $k = 1$, independent clusters of size 1, and a design for which $E\|n^{-1/2}\mathbf{X}^\top \mathbf{D}(\theta)\boldsymbol{\varepsilon}\|^2$ is bounded below by a positive multiple of $(\theta - \theta_0)^2$.

Exercise C2.

Let $\hat{\boldsymbol{\theta}}$ be independent of $\mathbf{Y}$ with $E\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}_0$, and write $g(\boldsymbol{\theta}) = \mathbf{a}^\top (\mathbf{X}^\top \mathbf{V}(\boldsymbol{\theta})^{-1} \mathbf{X})^{-1} \mathbf{a}$ for the reported variance of $\mathbf{a}^\top \hat{\boldsymbol{\beta}}_F$ at scale $\sigma^2 = 1$, and $h(\boldsymbol{\theta})$ for its true variance. Show that $h(\boldsymbol{\theta}) \geq g(\boldsymbol{\theta}_0)$ for every $\boldsymbol{\theta}$, with equality at $\boldsymbol{\theta}_0$, and deduce that $Eg(\hat{\boldsymbol{\theta}})$ understates $Eh(\hat{\boldsymbol{\theta}})$ whenever g is concave near $\boldsymbol{\theta}_0$. Why does this not settle the question in general?

Solution. For fixed $\boldsymbol{\theta}$ the statistic $\mathbf{a}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}(\boldsymbol{\theta})}$ is a linear unbiased estimator of $\mathbf{a}^\top \boldsymbol{\beta}$, so Corollary 7.2.3 gives $h(\boldsymbol{\theta}) \geq h(\boldsymbol{\theta}_0) = g(\boldsymbol{\theta}_0)$, with equality at $\boldsymbol{\theta}_0$. Conditionally on $\hat{\boldsymbol{\theta}}$ the report is $g(\hat{\boldsymbol{\theta}})$ and the truth is $h(\hat{\boldsymbol{\theta}}) \geq g(\boldsymbol{\theta}_0)$; concavity of g and Jensen's inequality give $Eg(\hat{\boldsymbol{\theta}}) \leq g(\boldsymbol{\theta}_0)$. In general g need not be concave and $\hat{\boldsymbol{\theta}}$ is not independent of $\mathbf{Y}$; that dependence is what [Example \(How many clusters are enough?\)](#) measures.